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# Dominant Psychosocial Response & Distress Classification in Oncology Messages

Final - methodology, results, and conclusions

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## 1

## PROJECT DEFINITION

# Task, recap and refinement

- Goal: from one oncology message, predict the dominant psychosocial response (7 classes) and the ordinal distress level (low / medium / high).
- Data: 2,033 generated messages, 2,025 valid audited, and 1,273 model-ready items across quality tiers - no real patient data, no label leakage.
- Methods & metrics: three model families compared with macro-F1, ordinal weighted kappa / MAE, and 5000-resample bootstrap, validated by a blind human reading.

Refinement since proposal: “emotion” -> “dominant psychosocial response”; added dual-judge auditing, quality tiers, the leakage ban, and ordinal-aware distress metrics.

## 2

## ACHIEVEMENTS &amp; NOVELTY

## What we built and what is new

- Built and audited a 2,025-message synthetic corpus (2,033 generated); 1,273 items retained as model-ready, in strict / silver / consensus tiers.
- Compared three model families on both tasks; DistilBERT was evaluated over five random seeds, while TF-IDF and zero-shot baselines were deterministic; with bootstrap significance testing and a blind human validation.
- Delivered a consolidated error analysis (anger artifact, anxiety over-prediction, low / medium distress boundary).

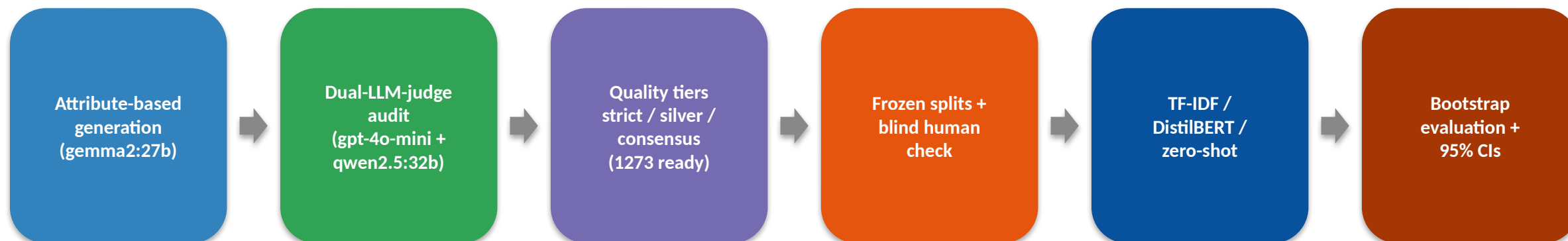
### Novelty

We jointly study fine-grained psychosocial response type and ordinal distress intensity using an attribute-controlled synthetic corpus, dual-LLM auditing, quality tiers, and blind human validation.

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METHODOLOGY

# End-to-end methodology



Effort for quality: five random seeds per DistilBERT configuration (report mean +/- std); TF-IDF and zero-shot baselines are deterministic. Plus a class-weighting control, 5000-resample bootstrap CIs, and a blind 67-item human check.

**Human check: response kappa 0.84; distress weighted kappa 0.80 linear / 0.86 quadratic.**

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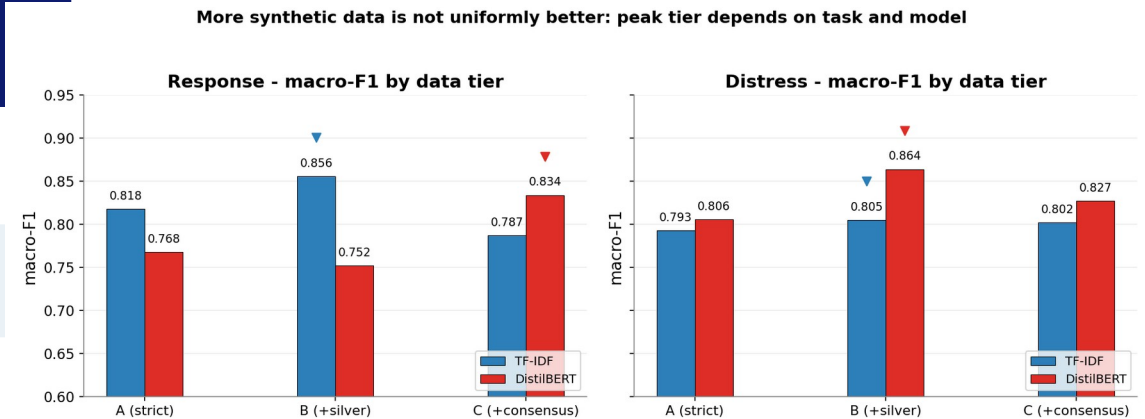
RESULTS

# Cross-task results

| Task     | Best lexical   | Best transf.       | Zero-shot  |
|----------|----------------|--------------------|------------|
| Response | TF-IDF-B 0.856 | DistilBERT-C 0.834 | BART 0.732 |
| Distress | TF-IDF-B 0.805 | DistilBERT-B 0.864 | BART 0.468 |

## Reliable conclusions (bootstrap):

- Supervised > zero-shot descriptively on both tasks; bootstrap-confirmed for TF-IDF-B on response and both models on distress.
- Lexical vs. transformer was NOT separable on 67 items (CIs include 0).
- More synthetic data is not uniformly better: peak tier depends on task and model.





## CONCLUSION

# Conclusions and future work

### Goals achieved

competitive classifiers for both tasks, with a rigorous, honestly-scoped evaluation.

### What we learned

a sparse lexical model is highly competitive for response; the transformer leads distress on the mean but not provably; more consensus data is not always better.

### Residual errors

concentrated at semantically ambiguous boundaries (low / medium distress), not extreme low/high errors; high distress was detected reliably on the synthetic test set.

### Scope

synthetic-only evaluation, one human annotator, and a 67-item test set.

### Future work

a multi-annotator gold standard, a more context-aware distress model, and validation on real (non-synthetic) text.